💡 **Useful Formulas for kinematics**

(1) $v = v_0 + at$

(2) $\Delta x = v_0 t + \frac{1}{2}at^2$

(3) $v^2 = {v_0}^2 + 2a\Delta x$

Q1: You are standing on a bridge that is $80\,\mathrm{m}$ high. You throw a rock perfectly horizontally at a speed of $15\,\mathrm{m/s}$. We use $g = -10\,\mathrm{m/s}^2$.

(a) What is its initial vertical velocity?

(b) Calculate the time it takes to hit the water below.

(c) Calculate the horizontal distance the rock traveled before hitting the water.

Q2: A snowboarder wants to slide off a perfectly flat horizontal ledge that is $45\,\mathrm{m}$ high. To win the competition, he must safely land on a specific snow pad that is exactly 60m horizontally away from the base of the ledge.

(a) Using the height of the ledge, calculate how long (t) the snowboarder will be in the air.

(b) What must be the snowboarder's initial horizontal speed as he leaves the ledge to land exactly on the target?

Q3: A golfer hits a golf ball with an initial velocity of 50m/s at an angle of 37° above the ground.

(a) Calculate the initial horizontal velocity (v_{0x}) and vertical velocity (v_{0y}).

(b) Calculate the maximum height (H) the golf ball reaches.

(c) Calculate the time that the golf ball takes to reach the highest point.

Q4: Based on the golf ball from Q3:

(a) Find the total time the ball is in the air.

(b) Calculate the total horizontal distance the golf ball traveled.

Q5: Based on the exact same golf ball from Q3 and Q4, and use the symmetry of projectiles:
(a) Right before it hits the ground, what is its horizontal velocity (v_x) and vertical velocity (v_y) respectively?
(b) Using your answers in (a) and the Pythagorean theorem, calculate the total speed of the ball right before it hits the ground.

Q6: A catapult fires a heavy stone with an initial velocity of $50\,\mathrm{m/s}$ at an angle of 53° above the ground. There is an enemy castle wall that is exactly $120\,\mathrm{m}$ away horizontally. The height of this castle wall is $60\,\mathrm{m}$.
(a) What is the horizontal and vertical velocity $v0x$ and $v0y$?
(b) Calculate how many seconds (t) it takes for the stone to travel $120\,\mathrm{m}$ horizontally.
(c) Calculate the height of the stone at that exact moment.
(d) Did the stone fly over the wall? Explain your reason.